



Text Mining I

Lecture 13

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Outline

- What is Text Mining
- Text Data Preprocessing
- Text Representation
 - Vector Space Model
 - Topic Model
- Case Study



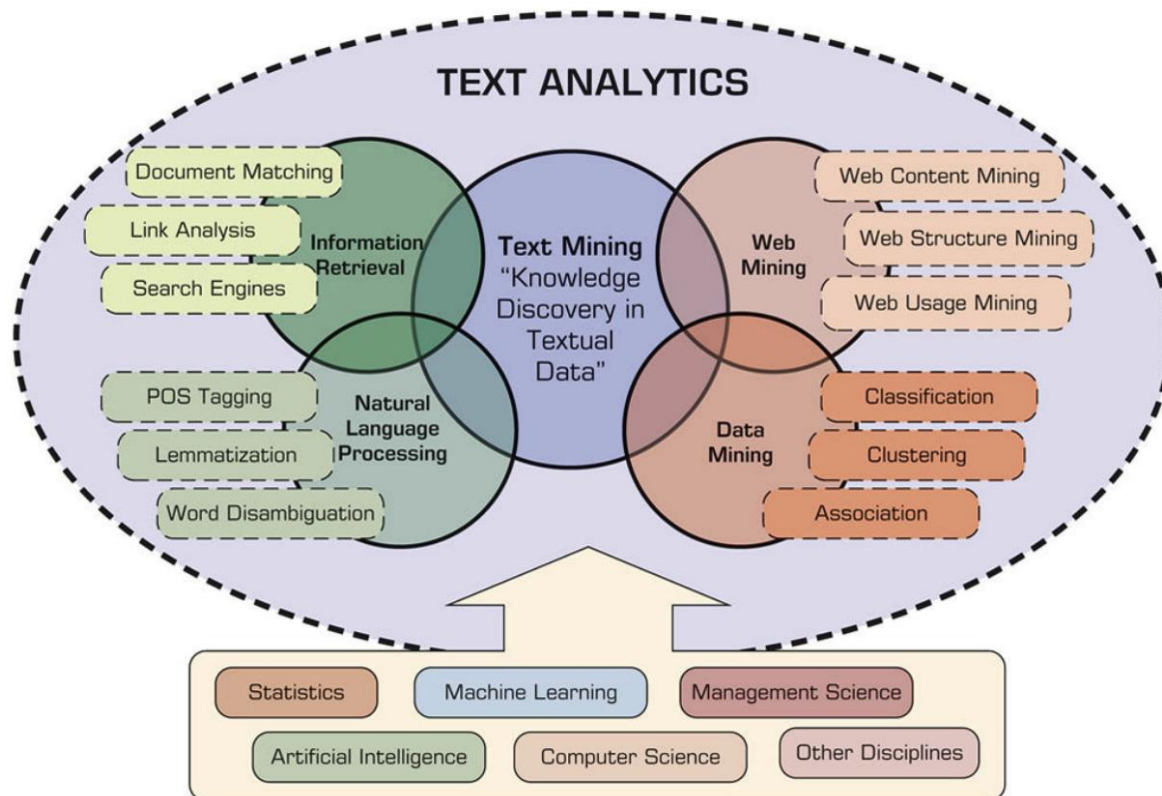
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Text Analytics

Text Analytics =

- Information Retrieval +
- NLP + Data Mining +
- Web Mining



Data Mining and Text Mining

Data Mining / Knowledge Discovery



Structured Data

HomeLoan (
Loanee: Frank Rizzo
Lender: MWF
Agency: Lake View
Amount: \$200,000
Term: 15 years
)

Multimedia



Free Text

Frank Rizzo bought his home from Lake View Real Estate in 1992.
He paid \$200,000 under a 15-year loan from MW Financial.

Hypertext

[Frank Rizzo](#)
Bought
[this home](#)
from [Lake View Real Estate](#)
In **1992**.
...

Rapid Growth of Text Information

WWW



Desktop



Email



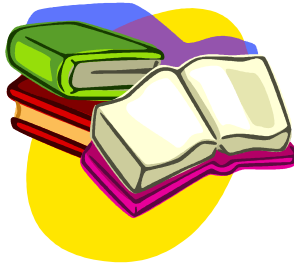
Intranet



Blog/Tweets



Literature

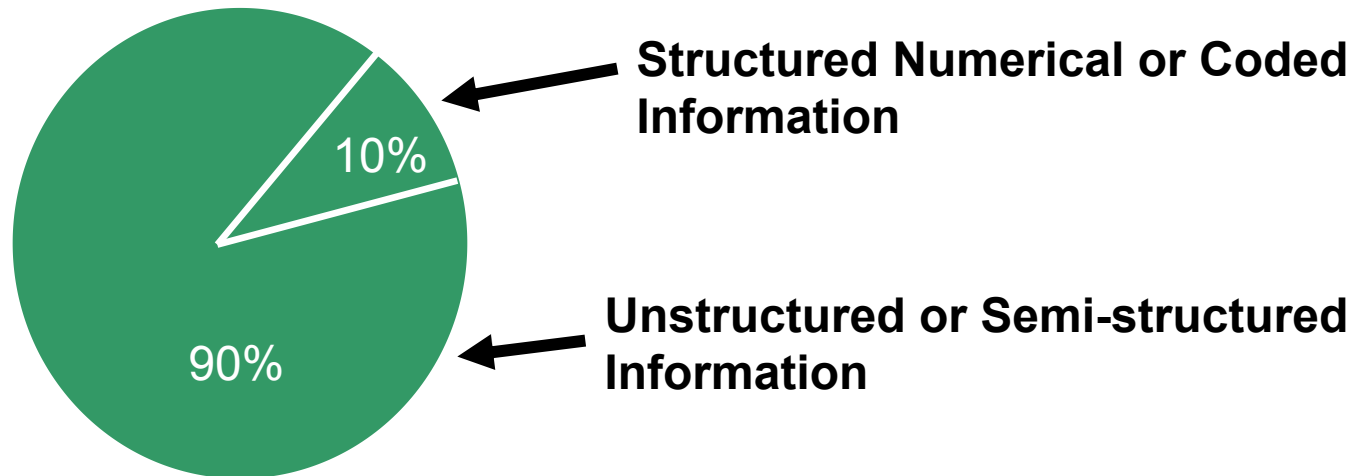


Online Review



Motivation for Text Mining

- Approximately **90%** of the world's data is held in unstructured formats (source: Oracle Corporation)
- Information intensive business processes demand that we transcend from simple document retrieval to “knowledge” discovery.

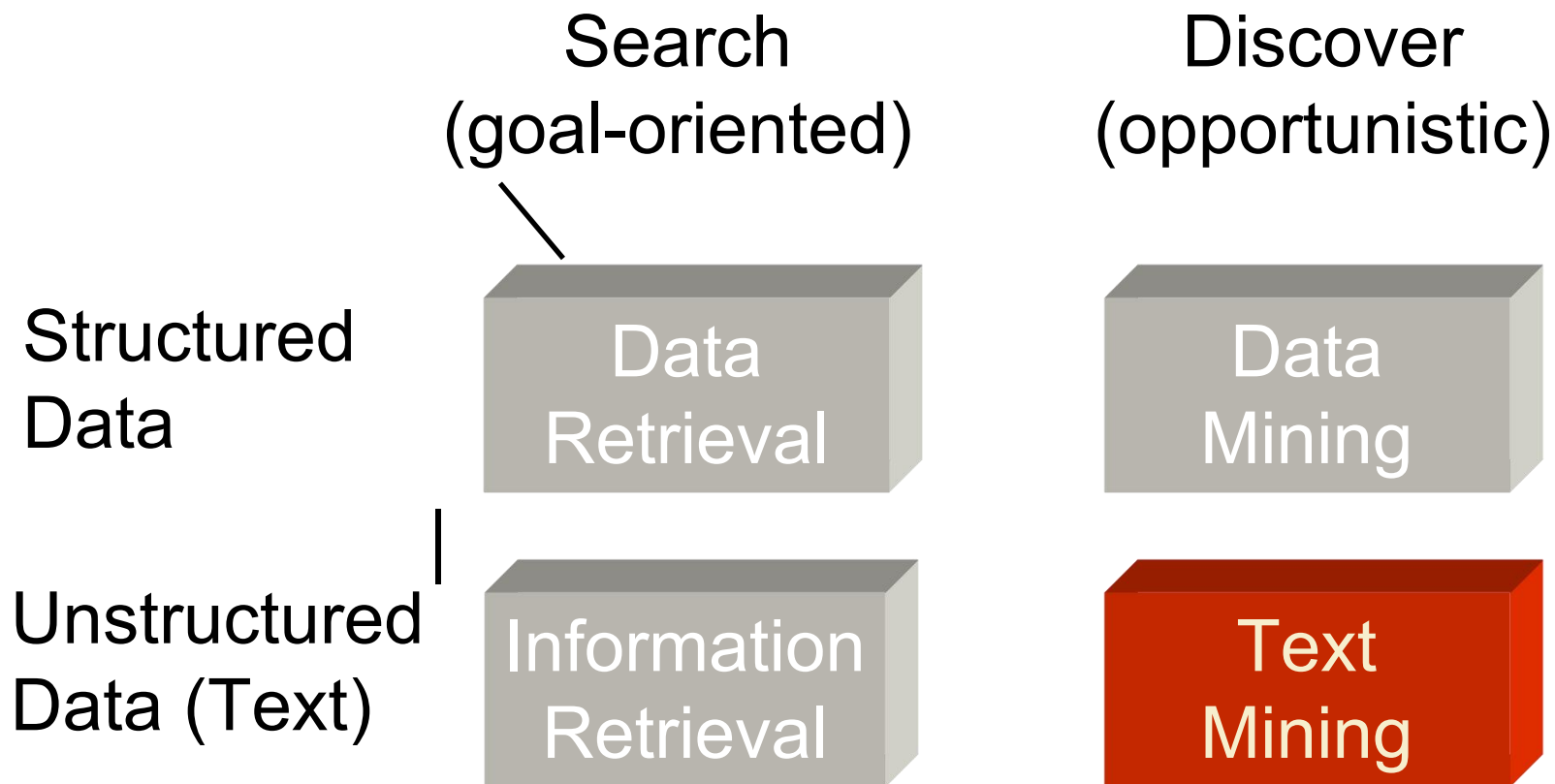




What is Text Mining?

- Also known as **Text Data Mining**
- **Process** of examining large collections of unstructured textual resources in order to generate new information
- **Explore patterns in textual data**
 - Find **latent topics**
 - Find **topical trends**
 - Find **hidden patterns**

Text Mining



Applications

- Enterprise Business Intelligence
- Healthcare/Medical Records
- National Security
- Scientific Discovery
- Sentiment Analysis Tools
- Natural Language Service
- Publishing
- Automated Ad Placement
- Information Access
- Social Media Monitoring



Challenges of Text Mining

- Very high number of possible “dimensions”
 - All possible word and phrase types in the language!!
- Unlike data mining:
 - records (= docs) are not **structurally identical**
 - records are not statistically **independent**
- Complex and subtle **relationships between concepts** in text
 - “AOL merges with Time-Warner”
 - “Time-Warner is bought by AOL”
- Ambiguity and **context** sensitivity
 - automobile = car = vehicle = Toyota
 - Apple (the company) or apple (the fruit)



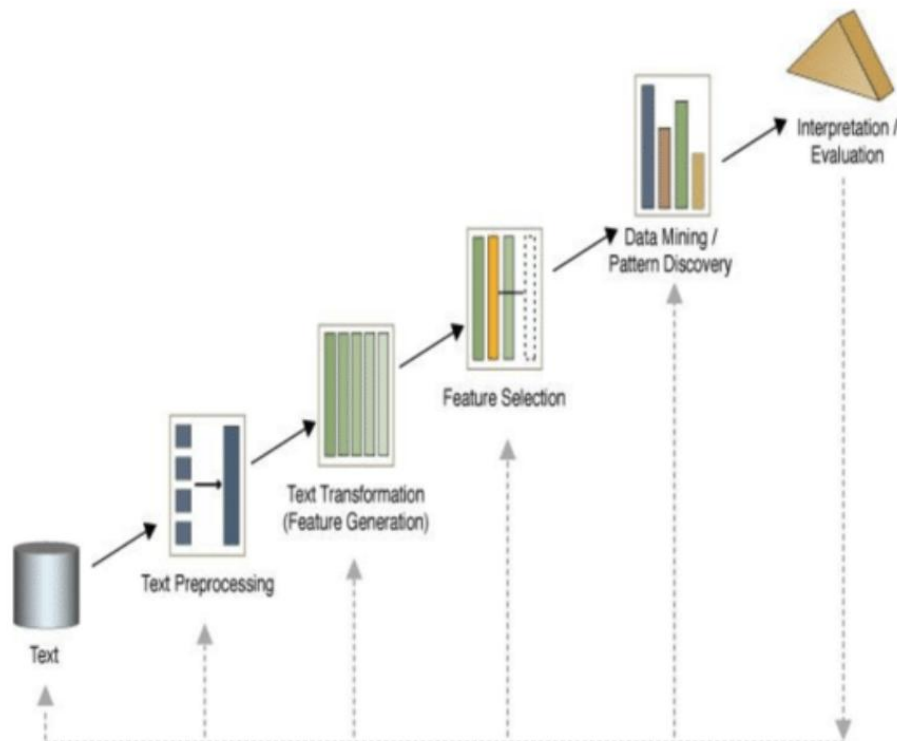
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Text Mining Process

Text mining process

- Text preprocessing
 - Syntactic/Semantic text analysis
- Features Generation
 - Bag of words
- Features Selection
 - Simple counting
 - Statistics
- Text/Data Mining
 - Classification-Supervised learning
 - Clustering-Unsupervised learning
- Analyzing results



Text Data Preprocess

Unstructured Data: Documents

Four score and seven years ago our fathers brought forth on this continent, **a new nation**, conceived in Liberty, and dedicated to the proposition that all men are created equal.

Now we are engaged in a great civil war, testing whether **that nation**, or ...

Data
Preprocess

Structural Data

nation – 5
civil - 1
war – 2
men – 2
died – 4
people – 5
Liberty – 1
God – 1
...

Text Preprocessing

- Text Cleanup
 - Normalize texts converted from binary formats (programs, media, images, and most compressed files)
 - Deal with tables, figures, and formulas
- Tokenization
 - Process of breaking a stream of text up into words, phrases, symbols, or other meaningful elements called tokens

Input: Friends, Romans, Countrymen, lend me your ears;

Output:

Friends	Romans	Countrymen	lend	me	your	ears
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Word Segmentation

- In some languages (e.g., Chinese), text is written without any spaces between words
信息检索和Web搜索是一门很有意思的课程。
- Word segmentation methods
 - Use a large vocabulary and take the longest vocabulary match
 - Machine learning sequence models (e.g., Markov models, conditional random fields)
 - Character k-grams

Stop Words

- Some extremely common words that would appear to be of little value in helping select documents matching a user need are excluded from the vocabulary entirely
 - Determined by collection frequency – the total number of times each term appears in the document collection
- Using a stop list significantly reduces the number of postings that a system has to store

a	an	and	are	as	at	be	by	for	from
has	he	in	is	it	its	of	on	that	the
to	was	were	will	with					

Stemming and Lemmatization

- How can we know “organize”, “organizes”, and “organizing” should map to the same word?
- Stemming and lemmatization: reduce inflectional forms and sometimes derivationally related forms of a word to a common base form
 - am, are, is → be
 - car, cars, car’s, cars’ → car
 - “the boy’s cars are different colors” → “the boy car be different color”

An Example

- **Input:**

- Document 1: Text mining is to identify useful information.
- Document 2: Useful information is mined from text.
- Document 3: Text mining is a subset of data mining.

- **Output:**

- Document 1: text, mine, identify, useful, information
- Document 2: useful, information, mine, text
- Document 3: text, mine, subset, data, mine



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Text Representation

- Text document is represented by a specific representation
 - i.e., the words (features) it contains and their occurrences
- Approach to generate attributes/document representation:
 - **Vector space model (i.e., Set/Bag of words)**
 - Topic model
 - Word embedding



Vector Space Model

- Vector representation doesn't consider the ordering of words in a document
 - i.e., “John is quicker than Mary”
“Mary is quicker than John”
- The words (features) and their occurrences are used

Bool Method

- Set of words

	text	mine	identify	useful	information	subset	data
Doc 1	1	1	1	1	1	0	0
Doc 2	1	1	0	1	1	0	0
Doc 3	1	1	0	0	0	1	1

Occurrences of a term

- Bag of words
- Use the number of occurrences of a term in a document
- Each document is a count vector

	text	mine	identify	useful	information	subset	data
Doc 1	1	1	1	1	1	0	0
Doc 2	1	1	0	1	1	0	0
Doc 3	1	2	0	0	0	1	1

Term Frequency

- TF: The term frequency $TF(t,d)$ of term t in document d is defined as the number of times that t occurs in d (i.e., $N(t,d)$).

	text	mine	identify	useful	information	subset	data	length
Doc 1	1	1	1	1	1	0	0	5
Doc 2	1	1	0	1	1	0	0	4
Doc 3	1	2	0	0	0	1	1	5

- $TF(t,d)$ can be normalized:

$$TF(t,d) = \frac{N(t,d)}{\sum_t N(t,d)}$$

$$TF(t,d) = a + (1 - a) \frac{N(t,d)}{\max_t N(t,d)}, \text{ if } N(t,d) > 0$$

- Rare terms are more informative than frequent terms
 - i.e., stop words
- Two documents containing a rare term are very likely to be relevant
 - i.e., 形而上学
- We want a high weight for rare terms
- We will use document frequency (DF)

- DF_t is the document frequency of t
- The number of documents that contain t
 - DF_t is an inverse measure of the informativeness of t
 - $DF_t \leq N$
- We define the IDF (inverse document frequency) of t by

$$IDF(t) = 1 + \log\left(\frac{N}{DF(t)}\right)$$

	text	mine	identify	useful	information	subset	data
Doc 1	1	1	1	1	1	0	0
Doc 2	1	1	0	1	1	0	0
Doc 3	1	2	0	0	0	1	1

TF-IDF weighting

- The TF-IDF weight of a term is the product of its TF weight and its IDF weight.

$$w(t, d) = TF(t, d) \times IDF(t)$$

- Best known weighting scheme in information retrieval
- Increases with the number of occurrences within a document
- Increases with the rarity of the term in the collection

$$w(t, d) = TF(t, d) \times IDF(t)$$

	text	mine	identify	useful	information	subset	data
Doc 1	1	1	2.099	1.405	1.405	0	0
Doc 2	1	1	0	1.405	1.405	0	0
Doc 3	1	2	0	0	0	2.099	2.099

Text Representation

- Text document is represented by a specific representation
 - i.e., the words (features) it contains and their occurrences
- Approach to generate attributes/document representation:
 - Vector space model (i.e., Set/Bag of words)
 - **Topic model**
 - Word embedding

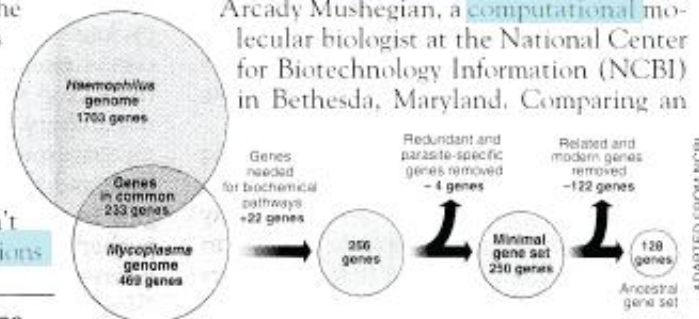
Topic Model

Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many **genes** does an **organism** need to **survive**? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for **life**. One research team, using **computer** analyses to compare known **genomes**, concluded that today's **organisms** can be sustained with just 250 genes, and that the earliest life forms required a mere 128 **genes**. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those **predictions**

"are not all that far apart," especially in comparison to the 75,000 **genes** in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a **genetic numbers game**, particularly as more and more **genomes** are completely mapped and sequenced. "It may be a way of organizing any newly **sequenced genome**," explains Arcady Mushegian, a **computational** molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an



* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

gene	0.04
dna	0.02
genetic	0.01
...	

life	0.02
evolve	0.01
organism	0.01
...	

data	0.02
number	0.02
computer	0.01
...	

SCIENCE • VOL. 272 • 24 MAY 1996

Topic Model

How can we write a document

1. Topics and Words

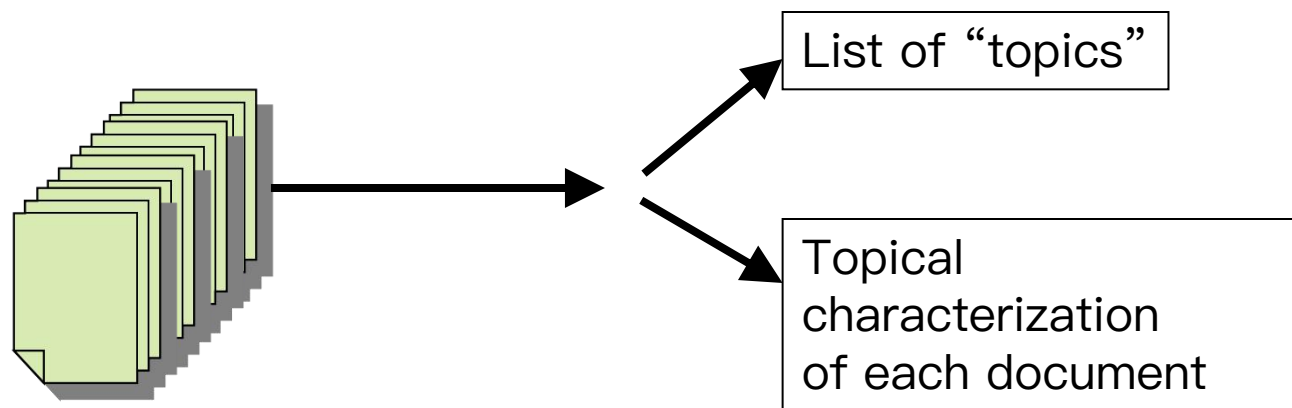
"Arts"	"Budgets"	"Children"	"Education"
NEW	MILLION	CHILDREN	SCHOOL
FILM	TAX	WOMEN	STUDENTS
SHOW	PROGRAM	PEOPLE	SCHOOLS
MUSIC	BUDGET	CHILD	EDUCATION
MOVIE	BILLION	YEARS	TEACHERS
PLAY	FEDERAL	FAMILIES	HIGH
MUSICAL	YEAR	WORK	PUBLIC
BEST	SPENDING	PARENTS	TEACHER
ACTOR	NEW	SAYS	BENNETT
FIRST	STATE	FAMILY	MANIGAT
YORK	PLAN	WELFARE	NAMPHY
OPERA	MONEY	MEN	STATE
THEATER	PROGRAMS	PERCENT	PRESIDENT
ACTRESS	GOVERNMENT	CARE	ELEMENTARY
LOVE	CONGRESS	LIFE	HAITI

How can we write a document

2. Writ the document

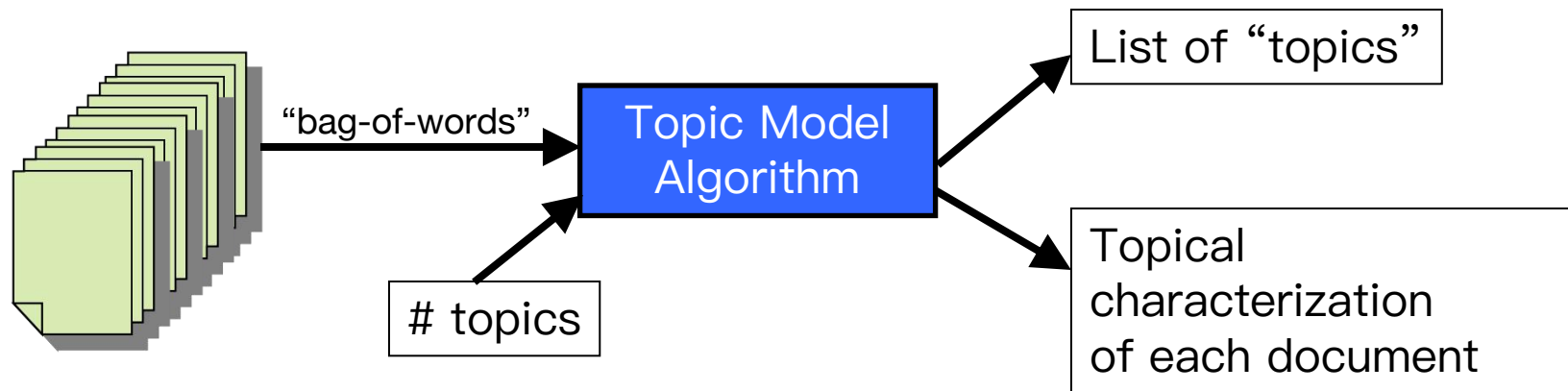
The William Randolph Hearst Foundation will give \$1.25 million to Lincoln Center, Metropolitan Opera Co., New York Philharmonic and Juilliard School. “Our board felt that we had a real opportunity to make a mark on the future of the performing arts with these grants an act every bit as important as our traditional areas of support in health, medical research, education and the social services,” Hearst Foundation President Randolph A. Hearst said Monday in announcing the grants. Lincoln Center’s share will be \$200,000 for its new building, which will house young artists and provide new public facilities. The Metropolitan Opera Co. and New York Philharmonic will receive \$400,000 each. The Juilliard School, where music and the performing arts are taught, will get \$250,000. The Hearst Foundation, a leading supporter of the Lincoln Center Consolidated Corporate Fund, will make its usual annual \$100,000 donation, too.

What we want to do



What is topic modeling?

- Learning “topics” from a set of documents in a statistical unsupervised fashion



- Many useful applications:
 - Improved web searching
 - Automatic indexing of digital historical archives
 - Specialized search browsers (e.g. medical applications)
 - Legal applications (e.g. email forensics)

Latent Dirichlet Allocation (LDA)

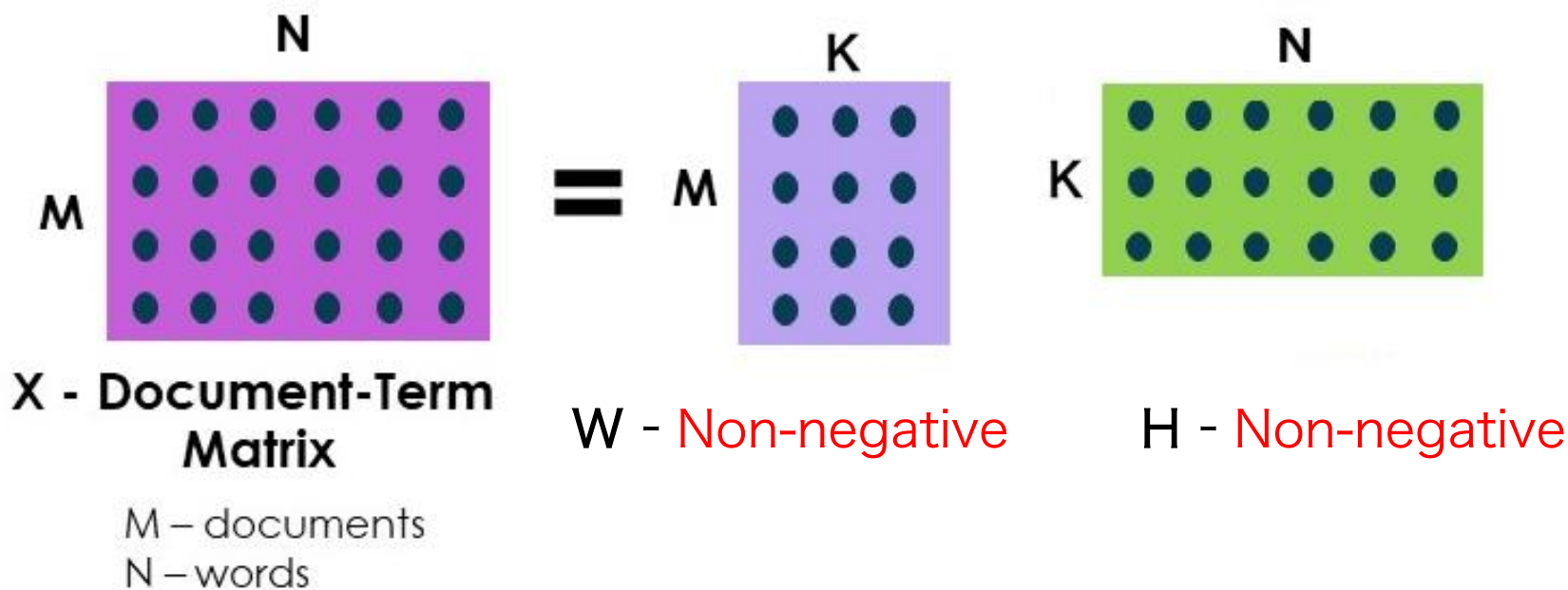
- History:
 - 1988: *Latent Semantic Analysis* (LSA)
 - Singular Value Decomposition (SVD) of word-document count matrix
 - 1999: *Probabilistic Latent Semantic Analysis* (PLSA)
 - Non-negative matrix factorization (NMF) -- version which minimizes KL divergence
 - 2003: *Latent Dirichlet Allocation* (LDA)
 - Bayesian version of PLSA

$$\begin{matrix} & D \\ W & \boxed{\text{P (word | doc)}} \end{matrix} \approx \begin{matrix} & K \\ W & \boxed{\text{P (word | topic)}} \end{matrix} * \begin{matrix} & D \\ K & \boxed{\text{P (topic | doc)}} \end{matrix}$$

- NMF

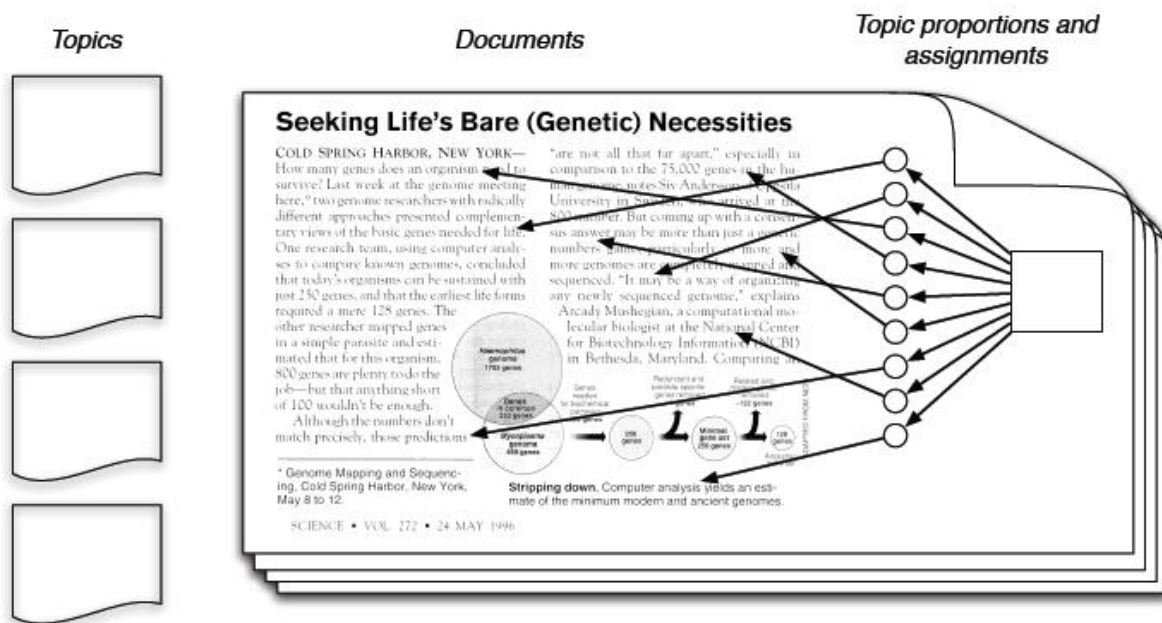
$$\hat{X} = WH$$

$$\min_{W,H} \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^n (X_{ij} - (WH)_{ij})^2, W_{ia} \geq 0, H_{bj} \geq 0$$



Latent Dirichlet Allocation

- Generative probabilistic model of a corpus
- Documents are represented as random mixtures over latent topics
- Topic is characterized by a distribution over words



Some Distributions

Binomial Distribution:

$$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Multinomial Distribution (多项分布):

$$P(x_1, x_2, \dots, x_k; n, p_1, p_2, \dots, p_k) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}$$

Some Distributions

Beta Distribution:

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

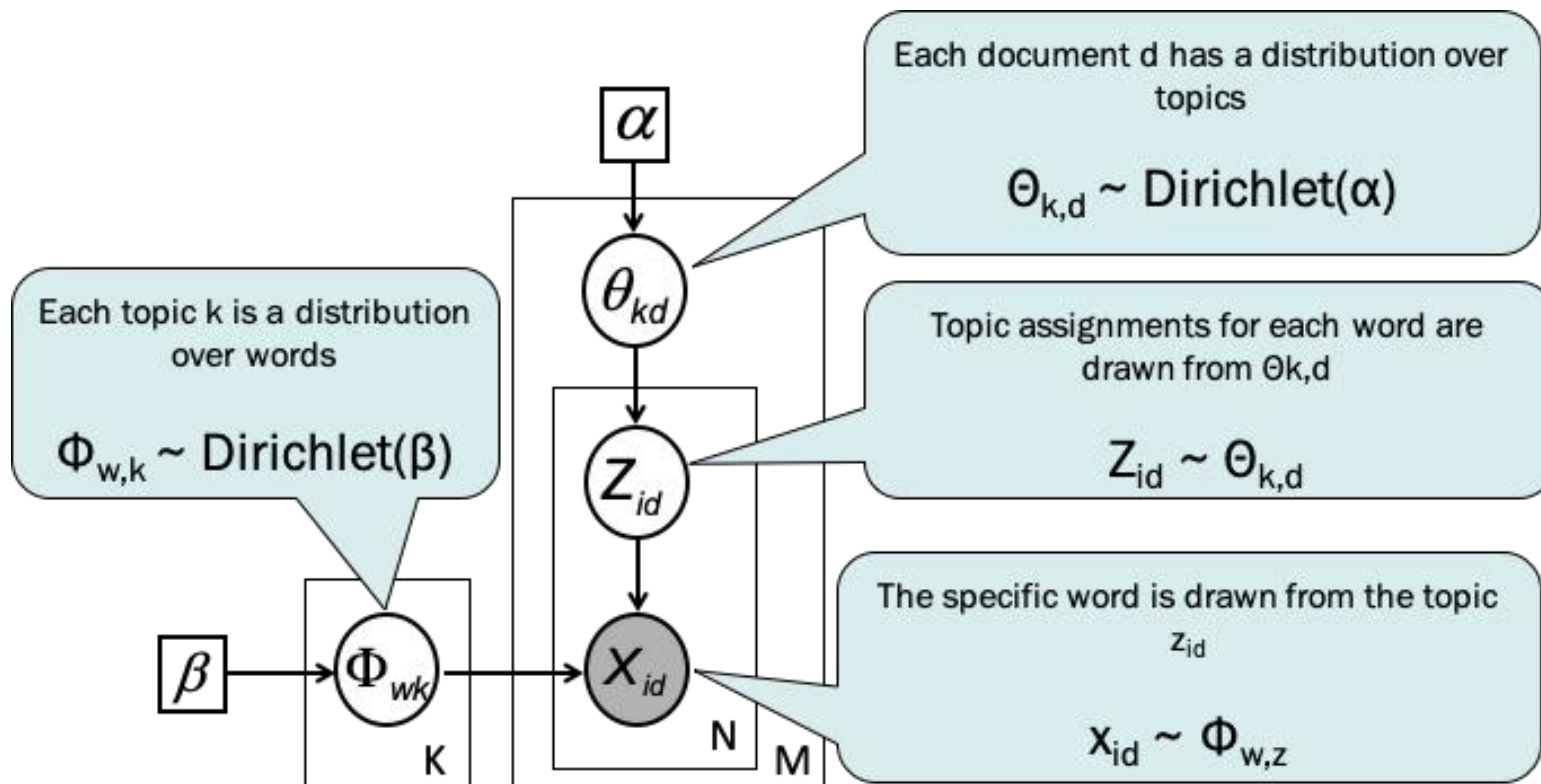
$$\frac{1}{B(\alpha, \beta)} = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)}$$

Dirichlet Distribution (狄利克雷):

$$f(x_1, x_2, \dots, x_k; \alpha_1, \alpha_2, \dots, \alpha_k) = \frac{1}{B(\alpha)} \prod_{i=1}^k x_i^{\alpha_i-1}$$

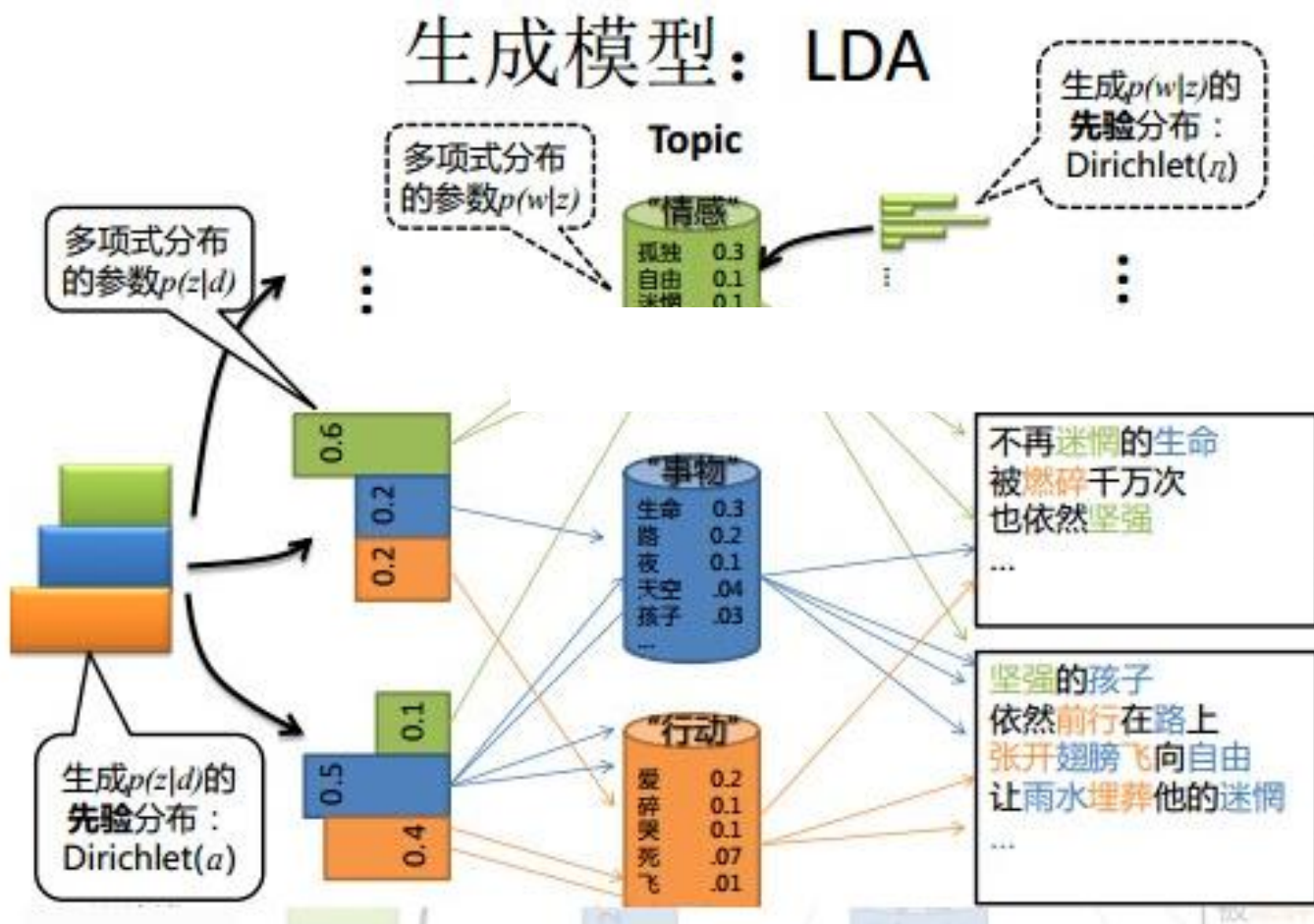
$$B(\alpha) = \frac{\prod_{i=1}^k \Gamma(\alpha_i)}{\Gamma(\sum_{i=1}^k \alpha_i)}, \sum_{i=1}^k x_i = 1$$

Graphical model for LDA

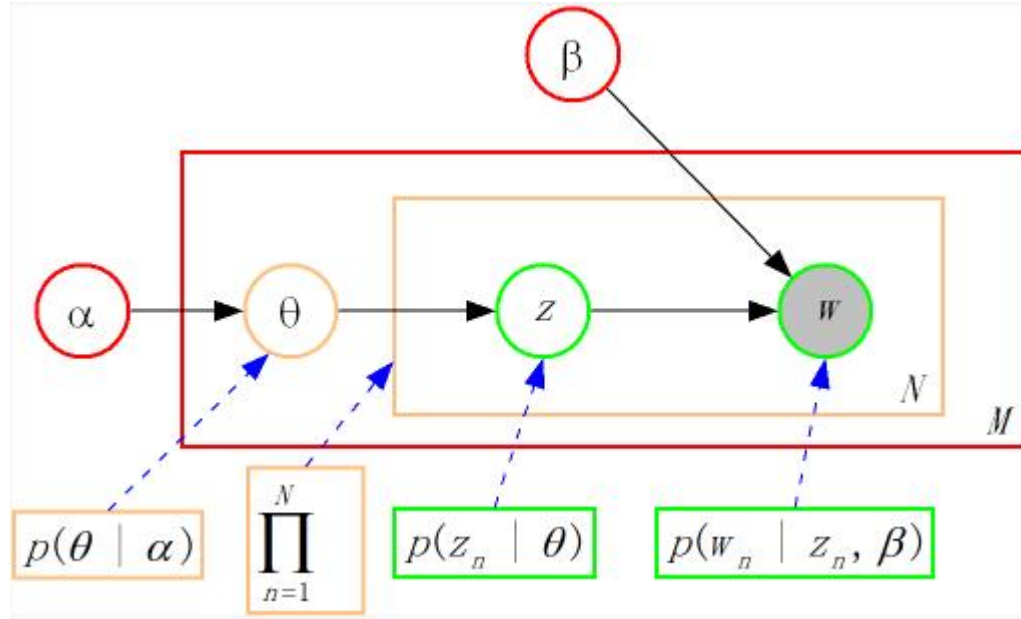


- Hidden/observed variables are in unshaded/shaded circles.
- Parameters are in boxes.
- Plates denote replication across indices.

Example



Text Mining and LDA



$$p(\theta, z, w | \alpha, \beta) = p(\theta | \alpha) \prod_{n=1}^N p(z_n | \theta) p(w_n | z_n, \beta)$$



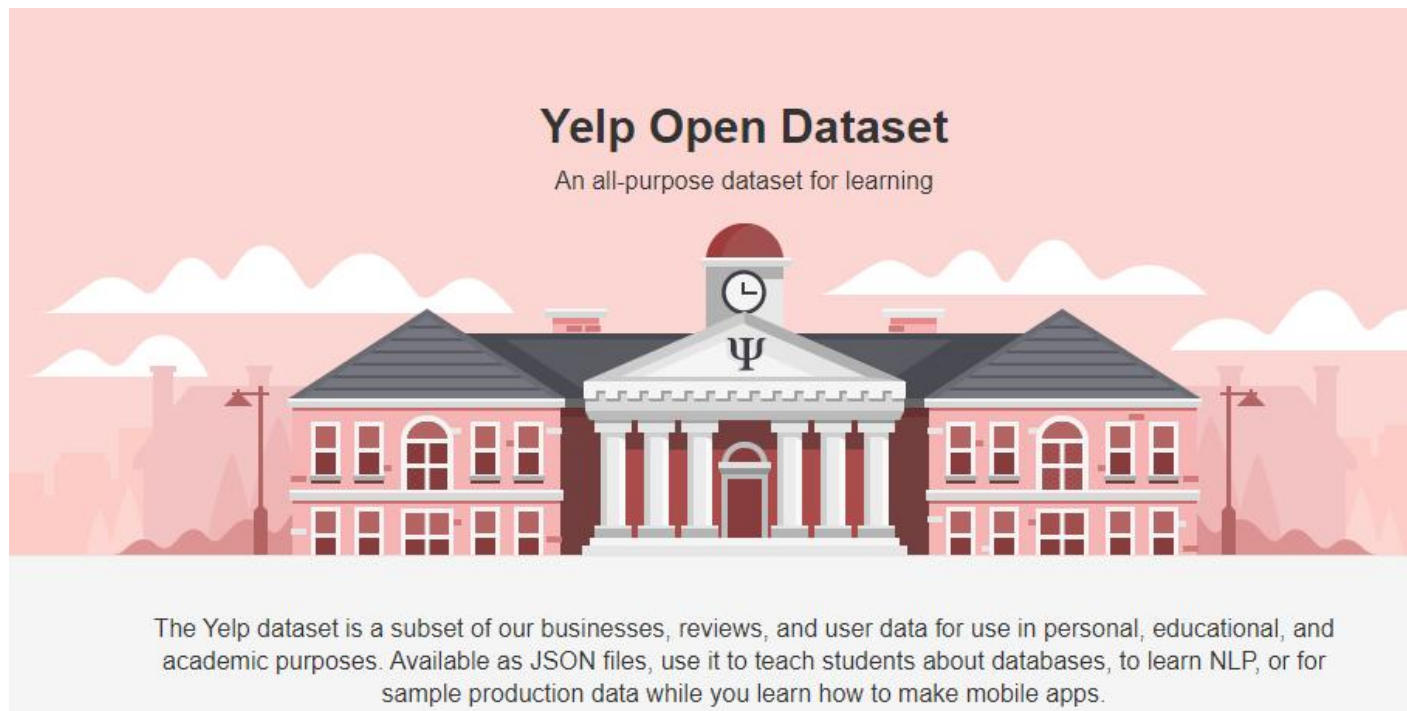
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 - Word embedding (next lecture)

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Case Study

- <https://www.yelp.com/dataset>



Case Study

- <https://www.yelp.com/dataset>

The Dataset



8,635,403 reviews



160,585 businesses



200,000 pictures



8 metropolitan areas

1,162,119 tips by 2,189,457 users

Over 1.2 million business attributes like hours, parking, availability, and ambiance

Aggregated check-ins over time for each of the 138,876 businesses

Case Study

- <https://www.yelp.com/dataset>

	review	sentiment
0	leave table feel ambivalent meh others would p...	0
1	time worth stick magical number three okay let...	0
2	love place boyfriend celebrate anniversary gla...	1
3	lovely even last night table great selection w...	1
4	table classic bistro good menu wine list cockt...	0

Case Study

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	review	sentiment
0	leave table feel ambivalent meh others would p...	0
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Some Takeaways

- Text data mining process
- TF-IDF
- LDA

The End

■ THANK YOU

Any questions?

